MATH 3080 Lab Project 4

Your Name Here

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```r
pi_norm <- function(x, conf.level = 0.95, alternative = "two.sided") {
  alpha <- 1 - conf.level
  n <- length(x)
  xbar <- mean(x)
  err <- sd(x) * sqrt(1 + 1 / n)
  crit <- switch(alternative,
                 "two.sided" = qt(alpha / 2, df = n - 1, lower.tail = FALSE),
                 "less" = -qt(alpha, df = n - 1, lower.tail = FALSE),
                 "greater" = qt(alpha, df = n - 1, lower.tail = FALSE),
                 # Below is the "default" switch, triggered if none of the above
                 stop("alternative must be one of two.sided, less, greater"))
  interval <- switch(alternative,
                     "two.sided" = c(xbar - crit * err, xbar + crit * err),
                     "less" = c(xbar + crit * err, Inf),
                     "greater" = c(-Inf, xbar + crit * err),
                     stop("How did I get here?"))
  attr(interval, "conf.level") <- conf.level
  interval
}
```

Problem 1

*The `cats` data set (**MASS**) contains the heart and body weight of a sample of male and female cats. Use the data set to estimate a 95% prediction interval for the body weight of a male cat. Assume that the body weight of cats is Normally distributed.*

```r
library(MASS)
head(cats)
```

```
##   Sex Bwt Hwt
## 1   F 2.0 7.0
## 2   F 2.0 7.4
## 3   F 2.0 9.5
## 4   F 2.1 7.2
## 5   F 2.1 7.3
## 6   F 2.1 7.6
```

```r
male_cats <- cats[cats$Sex == "M", ]


pi_norm(male_cats$Bwt)
```

```
## [1] 1.96728 3.83272
## attr(,"conf.level")
## [1] 0.95
```

Interpret what this prediction interval means with respect to the body weight of a male cat.

Response:

This interval tells us that there is a 95% chance that the next male cat in the sequence will have a bodyweight between 1.97kg and 3.83kg.

Problem 2

*The data set `SP500` (**MASS**) contains the returns of the S&P 500 stock index for the 1990s; that is, it's the ratio of the change of the index's price divided by the preceding day price. In principle, when predicting the direction of the stock market with the intention of buying stock, we are willing to be wrong in one direction but not another; we are okay with predicting the market grows too little and be pleasantly surprised than to predict the market grows more than it actually does. So compute a 99% lower prediction bound, assuming that stock returns are Normally distributed. (You should not trust this number. First the Normality assumption, despite being assumed a lot in finance, is not true. Second, stock returns are* not *an independent and identically distributed sample.)*

```r
head(SP500)
```

```
## [1] -0.2588908 -0.8650307 -0.9804139  0.4504321 -1.1856666 -0.6629097
```

```r
pi_norm(SP500, .99, "less")
```

```
## [1] -2.160704       Inf
## attr(,"conf.level")
## [1] 0.99
```

Problem 3

The data set ***abbey*** *(****MASS****) contains determinations of nickel content (ppm) in a Canadian syenite rock. The assumption of a Normal distribution clearly is inappropriate for this data set. Construct a 90% prediction interval for the next measurement from the data set. Use a nonparametric procedure.*

```r
no_param_pi <- function(x, conf.level = 0.95) {
  n <- length(x)
  x <- sort(x)
  j <- max(floor((n + 1) * (1 - conf.level) / 2), 1)
  conf.level <- (n + 1 - 2 * j)/(n + 1)
  interval <- c(x[j], x[n + 1 - j])
  attr(interval, "conf.level") <- conf.level
  interval
}


no_param_pi(abbey, .9)
```

```
## [1]   5.2 125.0
## attr(,"conf.level")
## [1] 0.9375
```

```r
head(abbey)
```

```
## [1] 5.2 6.5 6.9 7.0 7.0 7.0
```

Problem 4

Use the data from Problem 1 to construct a 95% tolerance interval for 99% of cats' body weight.

```r
library(tolerance)
```

```
## tolerance package, version 3.0.0, Released 2024-04-18
## This package is based upon work supported by the Chan Zuckerberg Initiative: Essential Open Source So
```

```r
normtol.int(cats$Bwt, alpha = 1 - 0.95, P = 0.99, side = 2)
```

```
##   alpha    P    x.bar 2-sided.lower 2-sided.upper
## 1  0.05 0.99 2.723611      1.332637      4.114585
```

Interpret what this tolerance interval means with respect to cat body weights.

There is a 95% chance that 99% of the data will be between 1.333 and 4.115kg.

Problem 5

The data set **`geyser`** *(**MASS**) contains both wait time between and duration of eruptions of the Old Faithful geyser in Yellowstone National Park. Use the data set to construct a nonparametric tolerance interval containing 90% of geyser eruptions (duration) with 99% confidence.*

```
nptol.int(geyser$duration, alpha = 1 - 0.99, P = 0.9, side = 2)
```

```
##   alpha   P 2-sided.lower 2-sided.upper
## 1  0.01 0.9      1.716667      4.966667
```

Interpret what this tolerance intervals means with respect to geyser eruptions.

Response

There is a 99% chance that 90% of all eruptions of old faithful have a duration between 1.717 seconds and 4.967 seconds.

Problem 6

*The data set `accdeaths` (**MASS**) contains a count of accidental deaths in the United States between 1973 and 1978.

Use this data set to construct a statistical interval for the mean number of accidental deaths over the last two years. i.e., calculate a replication interval for 1973-1976.

You will need to make a shortened version of the data for 1973-1976.

```
repint <- function(x, m = length(x), conf.level = 0.95,
                   alternative = "two.sided") {
  alpha <- 1 - conf.level
  n <- length(x)
  xbar <- mean(x)
  err <- sd(x) * sqrt(1 / m + 1 / n)
  crit <- switch(alternative,
                 "two.sided" = qt(alpha / 2, df = n - 1, lower.tail = FALSE),
                 "less" = -qt(alpha, df = n - 1, lower.tail = FALSE),
                 "greater" = qt(alpha, df = n - 1, lower.tail = FALSE),
                 # Below is the "default" switch, triggered if none of the above
                 stop("alternative must be one of two.sided, less, greater"))
  interval <- switch(alternative,
                     "two.sided" = c(xbar - crit * err, xbar + crit * err),
                     "less" = c(xbar + crit * err, Inf),
                     "greater" = c(-Inf, xbar + crit * err),
                     stop("How did I get here?"))
  attr(interval, "conf.level") <- conf.level
  interval
}
print(accdeaths)
```

```
##         Jan   Feb   Mar   Apr   May   Jun   Jul   Aug   Sep   Oct   Nov   Dec
## 1973  9007  8106  8928  9137 10017 10826 11317 10744  9713  9938  9161  8927
## 1974  7750  6981  8038  8422  8714  9512 10120  9823  8743  9129  8710  8680
## 1975  8162  7306  8124  7870  9387  9556 10093  9620  8285  8466  8160  8034
## 1976  7717  7461  7767  7925  8623  8945 10078  9179  8037  8488  7874  8647
## 1977  7792  6957  7726  8106  8890  9299 10625  9302  8314  8850  8265  8796
## 1978  7836  6892  7791  8192  9115  9434 10484  9827  9110  9070  8633  9240
```

```r
n = 24
repint(accdeaths[1:(length(accdeaths) - n)], n)
```

```
## [1] 8352.339 9324.619
## attr(,"conf.level")
## [1] 0.95
```

```r
max(accdeaths[48:72])
```

```
## [1] 10625
```

```r
min(accdeaths[48:72])
```

```
## [1] 6892
```

(Bonus +1 point) compare your interval to the observed mean over the years 1976-1978 and assess how well it did.

Our replication interval did not do very good. The actual minimum was 6892 and the actual maximum was 10625, compared to our 95% prediction interval of 8352 and 9324